

Now, we need to receive the connection from the client and send a message back, as shown below:

```
cli, a = serv.accept()
print(f"Connection established with {a}")
```

```
data = cli.recv(1024).decode()
print(f"Received from client: {data}")
cli.send("Hello I am your server!".encode())
```

Finally, we close the connection and the socket:

```
cli.close()
serv.close()
```

Save this code as *server.py* or something that signifies that this is the server code.

Let us now move to the client code. In the client code, too, the first step will be to import sockets and create a client socket:

```
import socket
cli = socket.socket()
```

The connection request is now sent to the server. Assuming that the device is in the same network, we give the IP address and the port number. In this case, since we are working on only one laptop, we give the same local host and the port number 8080 as we discussed above in the server section.

```
cli.connect(('127.0.0.1', 8080))
```

Once the connection is made, we need to send a message and receive the message from the server as an acknowledgement. This can be done using the lines of code given below:

```
cli.send("Hello I am the client!".encode())
data = cli.recv(1024).decode()
print(f"Received from server: {data}")
```

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
(base) Jishnus-MacBook-Air:desktop jishnusaurav$ python server.py
Hi I am the server, listening on port 8080...
```

Figure 1: Server listening for client calls

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
The default interactive shell is now zsh.
To update your account to use zsh, please run 'chsh -s /bin/zsh'.
For more details, please visit https://support.apple.com/kb/HT208050.
(base) Jishnus-MacBook-Air:~ jishnusaurav$ cd Desktop/
(base) Jishnus-MacBook-Air:Desktop jishnusaurav$ python client.py
Received from server: Hello I am your server!
(base) Jishnus-MacBook-Air:Desktop jishnusaurav$
```

Figure 2: Client final output

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
(base) Jishnus-MacBook-Air:desktop jishnusaurav$ python server.py
Hi I am the server, listening on port 8080...
Connection established with ('127.0.0.1', 58651)
Received from client: Hello from the client!
(base) Jishnus-MacBook-Air:desktop jishnusaurav$
```

Figure 3: Server final output

This will print the message received by the client as well.

Finally, we close the socket as shown below:

```
cli.close()
```

You can save this code as *client.py* or any name that will signify that this is the client code.

You should run the server code first. Just like there should be someone to listen to us before we start speaking, the client needs the server to be running first so that it can connect to it. When

you run the server code first, it will be waiting for a client's connection as shown in Figure 1.

Now you can run the client code on a different terminal/command prompt. The connection will be made and the message will be sent as shown in Figure 2. The server will also receive the message, and the output will be shown on the server terminal as shown in Figure 3.

This is just a beginning to socket programming. You can research further on its usage and application if you wish to. **END** 🐧

By: Jishnu Saurav Mittapalli

The author is a data scientist, interested in modern technologies like artificial intelligence and machine learning. He is currently working in the field of natural language processing.